

Malek Swilam

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EDUCATION

Oakton High School

Vienna, VA

Advanced Studies Diploma | Senior (Class of 2026) | GPA: 4.2 (Weighted) | SAT: 1510

- Relevant Coursework: AP Computer Science A, Advanced Data Structures & Algorithms, AP Computer Science Principles, AP Physics C: Electricity and Magnetism, AP Calculus BC, AP Statistics, AP Physics 1, STEM Advanced Engineering, Dual Enrollment Entrepreneurship

George Mason University – Honors College

Fairfax, VA

Incoming Computer Engineering, B.S.

Expected May 2030

EXPERIENCE

Alarm.com

Tysons, VA

Intern May 2025

- Supported quality engineering and software teams through device testing and lab operations for smart home IoT systems
- Configured and integrated IoT devices (locks, thermostats, cameras, contact/tilt sensors, doorbells, Flex I/O) for automated testing workflows; soldered components for sensor hardware
- Contributed to an AI project to assist lab technicians with diagnostics and workflow automation

ASSIP – George Mason University

Fairfax, VA (Hybrid)

Machine Learning Research Intern Summer 2025

- Built a full ML pipeline (data preprocessing, training, evaluation) for short-term rainfall forecasting by implementing and comparing LSTM, XGBoost, and ensemble models, evaluated using MAE, RMSE, and R^2
- Improved rainfall prediction accuracy by 12–15% over individual models by combining LSTM and XGBoost into an ensemble, achieving 96%+ of predictions within ± 1 mm of actual

FTC Robotics – Team 369 (Oakton)

Vienna, VA

Programming Lead | 2 Varsity Seasons 2023 – 2026

- Led all robot software development in Java across two seasons; mentored junior programmers in Java, Git, and FTC SDK over summer offseason
- Implemented motion profiling and trajectory planning via Road Runner library; sensor fusion of wheel odometry, IMU, and AprilTag vision with position and velocity PIDF controllers
- Led team to State Championship both seasons and World Championship qualification in one season

PROJECTS

OCRadar – Oral Cancer Detection App

Flutter, Dart, Azure AI Vision

- Built cross-platform iOS/Android app for real-time oral cancer screening via camera capture and Azure AI-powered lesion classification achieving 97% accuracy
- Integrated Azure AI Vision API for image analysis; designed an intuitive and polished UI for seamless user experience

SKILLS & HONORS

Languages: Java, Python, Dart, JavaScript

Frameworks & Tools: Flutter, PyTorch, OpenCV, Git/GitHub, Azure AI

Honors: First Tech Challenge World Championships (Ranked 8th in Division); Eagle Scout (Scouts BSA); T.R. Cook Award of Excellence in Technology, Optimist Club of Greater Vienna (2025)